

WRITE YOUR ANSWERS ON THIS EXAM PAPER

PLACE THIS PAPER AND ANY ANSWER BOOKLETS IN THE EXAM ENVELOPE PROVIDED

Name of Faculty: Natural, Mathematical & Engineering Sciences

Name of Department: Mathematics

Examination Period 2 (Summer 2026)

Candidate Number:		Seat Number (including colour zone if applicable)	
Module Code:	7CCMFM04		
Module Title:	STOCHASTIC ANALYSIS: OLD 2024-25 SYLLABUS		
Time Allowed:	TWO hours	Total Marks:	100
Total Pages:	20 pages	Total Questions:	4
Casio Fx83 or Fx85 Calculators Permitted	No		
Supplementary Booklet Required	This exam does not require a supplementary booklet.		
Any Other Permitted Materials	Books, notes or other written material may NOT be brought into this examination.		
Instructions for candidates	This paper consists of 4 questions; you must answer all the questions in the designated answer space. All questions carry equal marks. FULL MARKS will be awarded for complete answers to all FOUR questions. Write in PEN on THIS exam paper.		
This paper is part of an examination of the College counting towards the award of a degree. Examinations are governed by the College Regulations under the authority of the Academic Board.			
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If you require extra space to write your answers, please use the space at the end of the booklet on pages 18-21 (clearly write the question number on the extra page and the phrase 'see Extra space Page No. XXX' in the main section).

Q 1. Let $(W_t)_{t \geq 0}$ denote a standard Brownian motion on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $(\mathcal{F}_t)_{t \geq 0}$

- i) For an $(\mathcal{F}_t^W)_{t \geq 0}$ -stopping time τ , give the definition of $\mathcal{F}_\tau^W$. (proof not required). **[30%]**

Solution. $\mathcal{F}_\tau^W$ is the set of events A such that $A \cap \{\tau \leq t\} \in \mathcal{F}_t$

- ii) Let

$$X_t = \int_0^t \kappa(u, t) dW_u$$

where $\kappa \in L^2$. Compute $\mathbb{E}(X_t)$ and write out an integral formula for $\mathbb{E}(X_t^2)$ (you do not need to evaluate this integral). **[40%]**

Solution.

$$\begin{aligned} \mathbb{E}(X_t) &= 0 \\ \mathbb{E}(X_t^2) &= \int_0^t \kappa(t-u)^2 du. \end{aligned}$$

where the second expression follows from the general Ito isometry.

- iii)

Q 2. Let $(W_t)_{t \geq 0}$ denote a standard Brownian motion on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $(\mathcal{F}_t)_{t \geq 0}$.

- i) What is the conditional distribution of $W_t | W_{s_1}, \dots, W_{s_n}$ where $s_1 < \dots < s_n < t$. **[30%]**

Solution. $W_t | W_{s_1}, \dots, W_{s_n} \sim N(W_{s_n}, t - s_n)$

- ii) Let $B_t^{(i)} = (1-t)W_{\frac{t}{1-t}}^{(i)}$ for $t \in [0, 1)$, where $(W^{(i)})_{i=1..3}$ are three independent Brownian motions, and set $R_t = \sqrt{\sum_{i=1}^3 (B_t^{(i)})^2}$. Compute $\mathbb{E}(R_t^2)$. What can we say about the process R_t as $t \rightarrow 1$? **[40%]**

Solution.

$$\mathbb{E}(R_t^2) = \mathbb{E}\left(\sum_{i=1}^3 (B_t^{(i)})^2\right) = 3(1-t)^2 \frac{t}{1-t} = 3t(1-t).$$

Hence R_t is a non-negative process on $[0, 1)$ with $R_0 = 0$ and $\lim_{t \rightarrow 1} \mathbb{E}(R_t^2) = 0$, so in particular $R_t \rightarrow 0$ as $t \rightarrow 1$.

- iii) Let $X_t = \mu t + \sigma W_t$ and $\bar{X}_t = \max_{0 \leq s \leq t} X_s$. Using that

$$\mathbb{P}(\bar{X}_t > b | X_t = x) = h(b) = e^{-\frac{2b(b-x)}{\sigma^2 t}}$$

for $b \geq \max(x, 0)$, explain how you would simulate $\bar{X}_t$ given that $X_t = x$. **[30%]**

Solution. Set $\bar{X}_t = h^{-1}(U)$ where $U \sim U[0, 1]$.

Q 3. Throughout this question, we let $W = (W_t)_{t \geq 0}$ be a real-valued Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and let $M_t = \max_{0 \leq s \leq t} W_s$ denote the running maximum process of W and $\tau_b = \inf\{t \geq 0 : W_t = b\}$.

- i) State the reflection principle for Brownian motion and use it to compute $\mathbb{P}(M_t \geq b, W_t \in dx)$. **[40%]**

Solution.

$$\mathbb{P}(M_t \geq b, W_t \leq x) = \mathbb{P}(W_t \geq 2b - x) = \Phi^c\left(\frac{2b - x}{\sqrt{t}}\right)$$

for $b \geq x$. Then

$$\mathbb{P}(M_t \geq b, W_t \in dx) = n\left(\frac{2b - x}{\sqrt{t}}\right)dx$$

where $n(\cdot)$ is the standard Normal density.

- ii) If $M_t = a$ for $t \in [t_1, t_2]$ and $M_t \neq a$ otherwise, what can we say about the $(\tau_b)_{b \geq 0}$ process at $b = a$? (hint: draw a picture). **[30%]**

Solution. $(\tau_b)_{b \geq 0}$ has a jump at $b = a$ of size $t_2 - t_1$.

- iii) Write down an expression for the density of τ_b .

Solution. See Reflection Principle chapter

[30%]

Q 4. Let $(W_t)_{t \geq 0}$ denote a standard Brownian motion on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $(\mathcal{F}_t)_{t \geq 0}$.

i) Consider the SDE

$$dX_t = X_t^2 dW_t$$

with $X_0 > 0$. Derive the SDE satisfied by $Y_t = \frac{1}{X_t}$, in terms of Y_t . **[30%]**

Solution. From Ito's lemma,

$$dY_t = \frac{1}{Y_t} dt - dW_t$$

ii) Let S satisfy the SDE

$$dS_t = \sigma S_t^\beta dW_t$$

with $S_0 > 0$ for $\beta > 0$ (with absorption at zero), and let $\mathcal{F}^S$ denote the natural filtration of S . Using that a general one-dimensional diffusion of the form $dX_t = \sigma(X_t)dW_t$ is an $\mathcal{F}^X$ -martingale if and only if $\int_r^\infty xm(x)dx = \infty$ for all $r > 0$ (where $m(x) = 1/\sigma(x)^2$), compute the range of β values for which S is an $\mathcal{F}^S$ -martingale. **[40%]**

Solution.

$$\int_r^\infty xm(x)dx = \int_r^\infty \frac{x}{x^{2\beta}} dx = \frac{x^{2-2\beta}}{2-2\beta} \Big|_{x=r}^{x=\infty}. \quad (1)$$

But the final expression is infinite if and only if $\beta \leq 1$, so S is a martingale if and only if $\beta \leq 1$.

Write your solutions to the four questions here.

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Solution.

$$\int_r^\infty xm(x)dx = \int_r^\infty \frac{x}{x^{2\beta}}dx = \frac{x^{2-2\beta}}{2-2\beta} \Big|_{x=r}^{x=\infty}. \quad (2)$$

But the final expression is infinite if and only if $\beta \leq 1$, so S is a martingale if and only if $\beta \leq 1$.

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